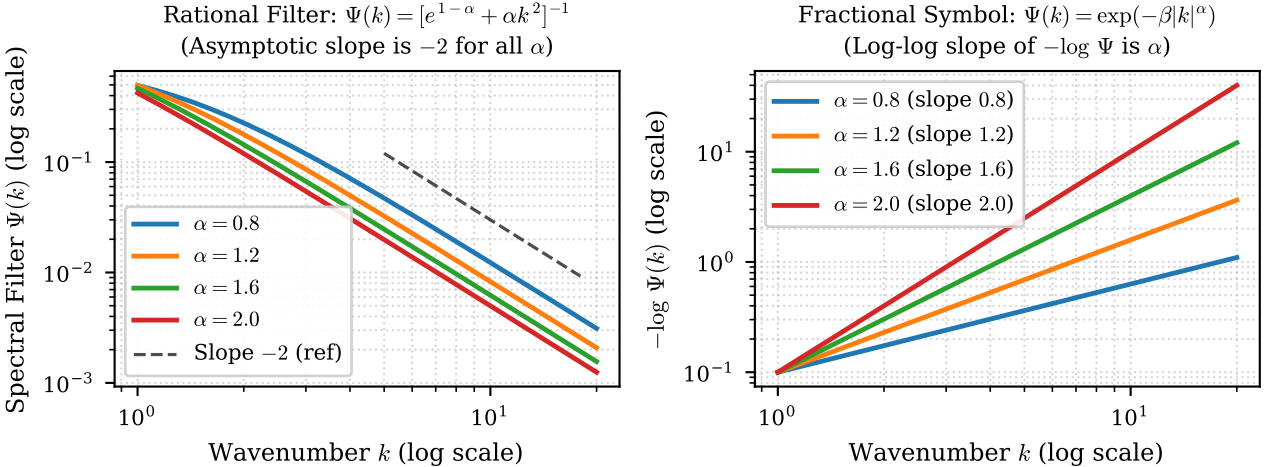
Rational Filter: $\Psi(k) = [e^{1-\alpha} + \alpha k^2]^{-1}$
(Asymptotic slope is -2 for all α)
Spectral Filter $\Psi(k)$ (log scale)
10^{-1}
10^{-2}
10^{-3}
10^{0}
10^{1}
Wavenumber k (log scale)
$\alpha = 0.8$
$\alpha = 1.2$
$\alpha = 1.6$
$\alpha = 2.0$
Slope -2 (ref)
Fractional Symbol: $\Psi(k) = \exp(-\beta|k|^{\alpha})$
(Log-log slope of $-\log \Psi$ is α)
$-\log \Psi(k)$ (log scale)
10^{-1}
10^{0}
10^{1}
10^{0}
10^{1}
Wavenumber k (log scale)
$\alpha = 0.8$ (slope 0.8)
$\alpha = 1.2$ (slope 1.2)
$\alpha = 1.6$ (slope 1.6)
$\alpha = 2.0$ (slope 2.0)